ANSWER KEY & MARKING SCHEME — SET A

Chapter 14: **Probability** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 0	No favourable outcome
2	**(a)** 0.65	$1 - 0.35$
3	**(a)** $\frac{1}{3}$	$\{5, 6\}$: $\frac{2}{6}$
4	**(a)** $\frac{1}{13}$	$\frac{4}{52}$
5	**(b)** Incorrect — $0 \le P(E) \le 1$	Probability is never negative
6	**(a)** $\frac{1}{4}$	Only TT out of 4
7	**(a)** Student A is correct	Only 2 is an even prime
8	**(c)** 115%	$115\% > 1$
9	**(a)** $\frac{6}{13}$	A, A, A, I, I, O: 6 of 13
10	**(b)** 1	Property of probability

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	(i) 2,3,5,7,11,13,17,19 ⇒ $\frac{8}{20} = \frac{2}{5}$ (ii) 5,10,15,20 ⇒ $\frac{4}{20} = \frac{1}{5}$	1+1
12	(i) $\frac{2}{52} = \frac{1}{26}$ (ii) $\frac{12}{52} = \frac{3}{13}$	1+1
13	False. $\frac{1}{2}$ is a theoretical probability; it does not guarantee 5 heads in 10 tosses (we may get 4, 6, …). It only means heads occur about half the time in a large number of tosses.	1+1
14	Total 180. (i) $\frac{100}{180} = \frac{5}{9}$ (ii) $\frac{170}{180} = \frac{17}{18}$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) (2,6),(3,5),(4,4),(5,3),(6,2): $\frac{5}{36}$ (ii) $\frac{6}{36} = \frac{1}{6}$ (iii) 6 outcomes: $\frac{1}{6}$	1+1+1
16	(i) 7 numbers ⇒ $\frac{7}{50}$ (ii) 1,4,9,16,25,36,49 ⇒ $\frac{7}{50}$ (iii) $16 + 12 - 4 = 24$ ⇒ $\frac{24}{50} = \frac{12}{25}$	1+1+1
17	(i) $\frac{132}{144} = \frac{11}{12}$ (ii) 143 pens left, 12 defective ⇒ $\frac{12}{143}$	2+1
	OR: $\frac{x}{5+x} = 2 \times \frac{5}{5+x} \Rightarrow x = 10$ blue balls	1+1+1
18	25 outcomes. (i) $\frac{5}{25} = \frac{1}{5}$ (ii) 8 outcomes ⇒ $\frac{8}{25}$ (iii) $1 - \frac{1}{5} = \frac{4}{5}$	1+1+1
	OR: (i) Green $= 16$, blue $= 8$ (ii) $\frac{16}{28} = \frac{4}{7}$	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{1}{8}$ (ii) $\frac{1}{2}$ (iii) $\frac{6}{8} = \frac{3}{4}$ (iv) 1 (sure event) (v) Wrong — 9 is not on the wheel, P(9) $= 0$	1+1+1+1+1
	OR: 46 cards left. (i) $\frac{6}{46} = \frac{3}{23}$ (ii) $\frac{26}{46} = \frac{13}{23}$ (iii) $\frac{20}{46} = \frac{10}{23}$ (iv) $\frac{2}{46} = \frac{1}{23}$	1+1+1+2
20	(i) 36 (ii) $\frac{1}{36}$, $\frac{6}{36} = \frac{1}{6}$ (iii) No — the 11 sums are not equally likely (iv) sums 2,3,5,7,11: $1 + 2 + 4 + 6 + 2 = 15$ ⇒ $\frac{15}{36} = \frac{5}{12}$	1+2+1+1
	OR: (i) $\frac{x}{12}$ (ii) $\frac{x+6}{18} = \frac{2x}{12} \Rightarrow x = 3$ (iii) $\frac{9}{18} = \frac{1}{2}$	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 14: **Probability** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1	All outcomes favourable
2	**(a)** $\frac{1}{3}$	$1-\frac{2}{3}$
3	**(a)** $\frac{1}{2}$	$\{1,3,5\}$
4	**(a)** $\frac{1}{26}$	$\frac{2}{52}$
5	**(b)** Incorrect; $P(\text{not } A)=0.3$	$P(A)+P(\bar{A})=1$
6	**(a)** $\frac{1}{2}$	HT, TH out of 4
7	**(a)** Student A is correct	Total outcomes are 10
8	**(d)** 0.99	$0 \le P \le 1$
9	**(a)** $\frac{2}{11}$	2 M's in 11 letters
10	**(a)** 1	Sure event

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	(i) $\frac{3}{8}$ (ii) $\frac{5}{8}$	1+1
12	(i) $\frac{6}{52}=\frac{3}{26}$ (ii) $\frac{1}{52}$	1+1
13	False. The two outcomes are not equally likely, so the classical formula cannot be used.	1+1
14	(i) HT, TH, TT ⇒ $\frac{3}{4}$ (ii) $\frac{1}{4}$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) (1,6),(6,1),(2,3),(3,2) ⇒ $\frac{4}{36}=\frac{1}{9}$ (ii) (4,6),(5,5),(6,4) ⇒ $\frac{1}{12}$ (iii) $\frac{9}{36}=\frac{1}{4}$	1+1+1
16	(i) $\frac{50}{100}=\frac{1}{2}$ (ii) 4, 9, …, 100: 9 ⇒ $\frac{9}{100}$ (iii) 2,3,5,7,11,13,17,19 ⇒ $\frac{8}{100}=\frac{2}{25}$	1+1+1
17	(i) $\frac{10}{35}=\frac{2}{7}$ (ii) $\frac{5}{7}$ (iii) $\frac{15}{40}=\frac{3}{8}$	1+1+1
	OR: White $=6$, black $=15$ ⇒ red $=30-21=9$	1+1+1
18	(i) $\frac{2}{6}=\frac{1}{3}$ (ii) $\frac{1}{6}$ (iii) A, E, A ⇒ $\frac{3}{6}=\frac{1}{2}$	1+1+1
	OR: (i) $\frac{4}{20}=\frac{1}{5}$ (ii) $\frac{4}{19}$	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{1}{4}$ (ii) $\frac{1}{52}$ (iii) $\frac{1}{26}$ (iv) $52-13-3=36$ ⇒ $\frac{9}{13}$ (v) $26+2=28$ ⇒ $\frac{7}{13}$	1+1+1+1+1
	OR: (i) $\frac{4}{36}=\frac{1}{9}$ (ii) $2+5+4+1=12$ ⇒ $\frac{1}{3}$ (iii) 8 outcomes ⇒ $\frac{2}{9}$ (iv) $3+2+1=6$ ⇒ $\frac{1}{6}$	1+1+1+2
20	(i) $\frac{6}{30}=\frac{1}{5}$ (ii) 1,4,9,16,25 ⇒ $\frac{1}{6}$ (iii) $6+5-1=10$ ⇒ $\frac{1}{3}$ (iv) 25 is counted twice	1+1+2+1
	OR: (i) $\frac{18-x}{18}$ (ii) $\frac{x+2}{20}=\frac{9}{8}\cdot\frac{x}{18}=\frac{x}{16}\Rightarrow 16x+32=20x\Rightarrow x=8$ (so P(not red) $=\frac{5}{9}$)	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.